Researchers achieve breakthrough in flexible electronics

Semiconductors are the very basic components of electronic devices, and these have improved our lives in many ways. These can be found in lighting, displays, solar modules and microprocessors that are installed in almost all modern-day devices, from mobile phones, washing machines and cars, to the emerging Internet of Things.

Dr Png Rui-Qi (left), Mervin Ang (middle) and Cindy Tang (right) working on conducting polymers that can provide unprecedented ohmic contacts for better performance in a wide range of organic semiconductor devices (Image courtesy: www.newelectronics.co.uk)

To make high-performance devices, however, good ohmic contacts with low electrical resistances are required to allow the maximum current to flow both ways between the electrode and the semiconductor layers. A team of scientists from National University of Singapore has successfully developed conducting polymer films that can provide unprecedented ohmic contacts to give superior performance in plastic electronics, including organic light-emitting diodes, solar cells and transistors.

The researchers discovered how to design polymer films with the desired extreme work functions needed to generally make ohmic contacts. Work function is the minimum amount of energy needed to liberate an electron from the film surface into vacuum. They showed that work functions as high as 5.8 electron-volts and as low as 3.0 electron-volts can now be attained for films that can be processed from solutions at low cost.

Flexible transparent conductor free of reflection and scattering

An effort is being made to search alternative transparent conductor materials that could replace indium-tin-oxide (ITO), especially for device flexibility. While the scientific community has investigated materials such as Al-doped ZnO, carbon nanotubes, metal nanowires, ultra-thin metals, conducting polymers and most recently graphene, none of

Flexible transparent conductor (Image courtesy: ICFO-The Institute of Photonic Sciences)

these have been able to present optimal properties that would make these replace ITO.

Ultra-thin metal films have been shown to present very low resistance, although their transmission is also low, unless anti-reflection undercoat and overcoat layers are added to the structure. ICFO-The Institute of Photonic Sciences researchers Rinu Abraham Maniyara, Vahagn K. Mkhitaryan, Tong Lai Chen and Dhriti Sundar Ghosh, under the guidance of Valerio Pruneri, ICREA Prof. at ICFO, have developed a room temperature processed multilayer transparent conductor optimising the anti-reflection properties to obtain high optical transmissions and low losses, with large mechanical flexibility properties.

In their study, they applied an Al-doped ZnO overcoat and a TiO2 undercoat layer with precise thicknesses to a highly-conductive silver ultra-thin film. By using destructive interference, the researchers showed that the proposed multilayer structure could lead to an optical loss of approximately 1.6 per cent and an optical transmission greater than 98 per cent in the visible.

The study shows the potential that this multilayered structure could have in technologies that aim at efficient and flexible electronic and optoelectronic devices.

Indian students create disaster recovery and surveillance robot

Fourteen-year-olds Anuj Verma and Shlok Jhawar, students of Delhi Public School, Bengaluru, India, under the mentorship of National Instruments, have successfully created a four-wheeled, all-terrain, multi-purpose robot named, Recon Rover. In their quest to do something for the society, these budding engineers aspired and built the disaster recovery and surveillance robot, Recon Rover, that can be used for surveillance during disasters and natural calamities.

The robot incorporates data from various actuators and sensors to give users the best control and accurate data from areas struck by natural calamities. It uses ultrasonic sensors to automatically avoid obstacles and prevent crashes along with sensing the presence of humans stuck in disaster zones. It sends real-time images and data from the disaster site while being controlled remotely.